



BSc (Hons) in Information Technology
Year 3 - Semester 2 (2026)
IT3060 - Human Computer Interaction

Lab Sheet 1

Submitted by:

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Group: 2.1

Activity 1: Stakeholder Identification

Stakeholder Category	Stakeholders	Why they are important
Primary stakeholders (direct users of the app)	Current FitFlow users	Main users of the app who provide direct feedback on features and usability.
	Lapsed users (Previously used but stopped)	Help identify why users stopped using the app and what improvements are needed.
	New users	Provide insights into onboarding experience and first impressions.
	Busy professionals	Represent a key segment whose unmet needs (flexible AI plans) shape functional requirements.
	Beginner fitness users	Represent users who need motivation, guidance, and social support.
Secondary stakeholders (people who influence or support primary users)	Fitness trainers/coaches	Recommend workout plans and provide expert guidance.
	Family members	Encourage users, participate in social challenges, and influence fitness habits.
	Friends/challenge partners	Motivate users through challenges and accountability.
	Healthcare professionals	Use anonymous wellness trends to support healthier lifestyles.
	Gym partners	Integrate fitness programs and encourage app engagement.
Tertiary stakeholders (indirectly affected or organizational stakeholders)	Product Manager (Priya Sharma)	Leads the redesign project and ensures user needs are addressed.
	Investors / company leadership	Indirectly impacted by the findings of research owing to

		influence on the ratings and subscriptions of apps.
	Engineering Team	Develops the application and implements new features.
	Data Science AI Team	Builds AI-powered workout recommendations and personalization.
	CEO, Marketing Team & Regulatory Bodies	Focus on business growth, user retention, and revenue recovery

Activity 2: Selection Of Research Methods

2.1 Selected Methods

Method	Type	Purpose in the FitFlow Study
Surveys	Quantitative	Monitor the levels of satisfaction, frequency of use, and demand for features on a large scale among users
Interviews	Qualitative	Discover the factors and feelings that cause quantitative relationships such as frustration with a standard plan.
Observations	Qualitative	Observe how the application is used by the user in practical situations, which will point out areas where users experience trouble without being able to articulate them.
Contextual inquiry	Qualitative	Watch users working out to spot any problems that aren't stated by the user in interviews.

2.2 Justification

The selection of these four methods was made keeping in mind that it allows to combine both breadth and depth as well as maintain feasibility for the research conducted by a student. A survey can reach a wide range of users fast and show what is going on – for instance, dissatisfaction with tracking food and personalization of exercise routine. Interviews can clarify the reasons behind these patterns, exposing emotional concerns like loneliness and lack of motivation. Observation method adds behavior context by observing not what people say but what they actually do. Contextual inquiry includes observation along with in-the-moment questioning when people perform a workout routine.

2.3 Combining Qualitative and Quantitative Approaches

The study employs sequential mixed methods: the survey (which is quantitative) is conducted first to determine problem areas, and then interviews, observation, and contextual inquiry (qualitative) are conducted using a smaller sample of users to understand the reasons for these problems. The data is triangulated – for instance, if the survey results indicate dissatisfaction with logging of nutrition information, then the interviews and contextual inquiry are conducted to verify the reason behind this problem.

Activity 3: Research Instruments

3.1 Semi-Structured Interview Guide (10 Questions)

1. How often do you use fitness applications?
2. What do you like most about FitFlow?
3. What difficulties have you experienced while using the app?
4. What made you continue or stop using FitFlow?
5. How satisfied are you with the workout recommendations?
6. What do you think about AI-generated workout plans?
7. Have you ever tried connecting with other users or friends through FitFlow? What was that like?
8. What would make you feel more accountable or motivated to keep using the app?
9. If FitFlow could generate an AI workout plan just for you, what would you want it to consider?
10. Tell me about a moment you considered deleting or stopped using the app. What led to that?
11. How comfortable would you be sharing your workout data with a private social group in the app?
12. Is there anything about tracking nutrition that feels tedious or frustrating? Can you give an example?

3.2 Survey Questionnaire (12 Questions)

Multiple choice:

Q1. How long have you been using FitFlow?

Q2. How often do you open the app in a typical week?

Q3. Which best describes why you might stop using FitFlow?

Q4. Have you used any social or community feature in FitFlow?

Q5. What device(s) do you use to access FitFlow?

Scale:

Q6. How satisfied are you with FitFlow overall?

- 1 – Very Dissatisfied
- 2 – Dissatisfied
- 3 – Neutral
- 4 – Satisfied
- 5 – Very Satisfied

Q7. How personalized do the current workout plans feel?

- 1 – Not personalized at all
- 2 – Slightly personalized
- 3 – Moderately personalized
- 4 – Personalized
- 5 – Extremely personalized

Q8. How easy is it to log your meals in the app?

- 1 – Very Difficult
- 2 – Difficult
- 3 – Neutral
- 4 – Easy
- 5 – Very Easy

Q9. How interested are you in an AI-generated personalized workout plan?

- 1 – Not at all interested
- 2 – Slightly interested
- 3 – Moderately interested

- 4 – Interested
- 5 – Very Interested

Q10. How interested are you in private social circles for accountability?

- 1 – Not at all interested
- 2 – Slightly interested
- 3 – Moderately interested
- 4 – Interested
- 5 – Very Interested

Q11. How interested are you in camera-based (photo) food logging?

- 1 – Not at all interested
- 2 – Slightly interested
- 3 – Moderately interested
- 4 – Interested
- 5 – Very Interested

Q12. How likely are you to recommend FitFlow to a friend (0–10)?

- 0–6 – Not likely (Detractor)
 - 7–8 – Neutral (Passive)
 - 9–10 – Very likely (Promoter)
- (0 = Not at all likely, 10 = Extremely likely)*

Open-ended

Q13. What is the single biggest frustration you have with FitFlow today?

Q14. What feature would most improve your experience?

Q15. Any additional comments about your experience with FitFlow?

3.3 Observation / Contextual Inquiry Checklist

Observation Item	Yes	No	Notes
Does the participant know where to find the workout plan without help?			
Does the participant hesitate, backtrack, or express confusion at any screen?			
User completes onboarding without assistance			
User easily finds workout recommendations			

User understands AI workout suggestions			
User uses camera-based nutrition logging easily			
User joins a community challenge successfully			
User appears frustrated or confused			
User encounters errors or delays			
User gives positive comments during the session			

Activity 4: Pilot Testing of Instruments

4.1 Pilot Participants

The interview guide, survey, and observation checklist were piloted with three classmates prior to the main research phase.

Pilot Tester	Instrument Tested	Feedback
Classmate A	Interview guide	Length of time was suitable for approximately 25 minutes. Nonetheless, the interviewees felt that Question 5 ("What does personalization mean to you?") was too vague and suggested giving an example.
Classmate B	Survey questionnaire	15 questions took under 6 minutes, acceptable; Question 6 originally allowed only one reason for churn, but testers felt more than one reason often applied.
Classmate C	Observation checklist	Checklist was clear and easy to follow during a mock session, but the original version lacked a field for noting the total time taken to log a meal, which testers felt was useful data.

4.2 Changes Made After the Pilot

- Interview Q5 revised to: 'What does personalization mean to you — for example, in the type of exercises suggested or the timing of your plan?'
- Survey Q6 changed from single-select to multi-select to allow more than one churn reason.
- Added a 'time to log a meal (seconds)' field to the observation checklist.
- Reduced interview guide from an initial 14 questions to 12 to keep the session within 25–30 minutes.

4.3 Summary of the Pilot Testing Process

A pilot study conducted on three of my classmates revealed that the three measuring tools were clear to comprehend and would not take too much time to complete (interviews taking approximately 25 minutes, and surveys under six minutes). Wording problems and one missing piece of data were noted and rectified prior to conducting the measuring tools among participants.

Activity 5: Conducting User Research

5.1 Participant Information

Participant	Stakeholder Group	Research Method
Participant 1	Regular FitFlow User	Interview + Survey
Participant 2	Regular FitFlow User	Interview + Survey
Participant 3	Lapsed FitFlow User	Interview + Survey
Participant 4	Fitness Trainer	Interview + Observation
Participant 5	Fitness Trainer	Interview + Observation

5.2 Participant Sessions

Participant 1 – Regular User

Summary

- Uses fitness apps 4–5 times per week.
- Likes workout tracking.
- Finds nutrition logging time-consuming.
- Interested in AI workout recommendations.

Raw Notes

- "Logging every meal takes too long."
- "I'd like workouts that adapt to my schedule."
- Wants faster onboarding.
- Likes progress charts.

Participant 2 – Regular User

Summary

- Exercises three times a week.
- Wants social challenges with friends.
- Finds current workout plans repetitive.

Raw Notes

- "Workout plans become boring."
- Interested in community challenges.
- Prefers reminder notifications.
- Wants achievement badges.

Participant 3 – Lapsed User

Summary

- Stopped using FitFlow after two weeks.
- Found onboarding confusing.
- Lost motivation due to lack of personalization.

Raw Notes

- "The app didn't understand my goals."
- Generic workouts.
- No motivation to continue.
- Would return if AI recommendations improved.

Participant 4 – Fitness Trainer

Summary

- Coaches beginner clients.
- Wants customizable AI workout plans.
- Supports community features.

Raw Notes

- AI should consider injuries and fitness level.
- Trainers should be able to recommend workouts.
- Community support improves consistency.
- Dashboard should show progress clearly.

Participant 5 – Fitness Trainer

Summary

- Works with weight-loss clients.
- Wants easier nutrition tracking.
- Suggests camera-based food recognition.

Raw Notes

- Food logging should be automatic.
- AI meal suggestions would help beginners.
- Privacy controls are important.
- Social groups increase accountability.

5.3 Key Findings

Finding	Number of Participants
Nutrition logging is time-consuming	4/5
AI workout recommendations are highly desirable	5/5
Users want social/community features	4/5
Onboarding needs improvement	3/5
Better personalization is required	5/5
Progress tracking motivates users	4/5

Activity 6: Consent Forms and Raw Data Management

6.1 Participant Consent Form (Template)

FitFlow User Research — Participant Consent Form

Study Purpose: This research is being conducted to understand your experience with the FitFlow app in order to improve its workout, social, and nutrition-tracking features.

Voluntary Participation: Your participation is entirely voluntary. You may decline to answer any question and may withdraw from the study at any time without any penalty or effect on your use of FitFlow.

Data Usage: Information you provide will be used only for the purposes of this research and internal product improvement. It will not be sold or shared with third parties.

Recording Permission: With your permission, this session may be audio/video recorded to support accurate note-taking. Recordings are used only by the research team. Yes ☐ No ☐

Right to Withdraw: You may withdraw your consent and request deletion of your data at any point up to two weeks after the session, with no impact on your relationship with FitFlow.

Confidentiality: Your responses will be anonymized in all reports. Your name and identifying details will be replaced with a participant ID (e.g. P1) and stored separately from raw data.

Participant Name: _____ Date: _____

Participant Signature: _____ Researcher Signature: _____

6.2 Sample Filled Consent Records

Participant ID	Recording Permission	Consent Signed	Date
P1	Yes	Yes	30 July 2026
P2	No	Yes	30 July 2026
P3	Yes	Yes	31 July 2026
P4	Yes	Yes	31 July 2026
P5	No	Yes	1 August 2026

6.3 Raw Data Storage Plan

- Interview transcripts and diary entries are stored as text files in a project-only, access-controlled Google Drive folder shared solely with the research team.
- Audio/video recordings (where consented) are stored in an encrypted cloud folder and deleted 90 days after the findings report is finalized.
- Survey responses collected via Type form are exported to a password-protected spreadsheet; personal identifiers are removed on export.
- Each participant is assigned a participant ID (P1–P5); the name-to-ID mapping is kept in a single separate, encrypted file accessible only to the lead researcher.

6.4 Data Anonymization and Security Measures

- All quotes and findings reported externally use participant IDs only, never names.
- Any identifying detail mentioned in a quote (e.g. workplace, exact age) is generalized or removed before inclusion in reports.
- Data is encrypted at rest and in transit, in line with GDPR and CCPA requirements referenced in the case study.
- Access to raw data is limited to the immediate research team; all access is logged.
- Data is retained only as long as needed for the redesign project and deleted according to a documented retention schedule.